Lab Exercise 15- Creating ConfigMaps in Kubernetes

In this lab exercise, you will learn how to create ConfigMaps in Kubernetes using different methods: from literal values, from a file, from an environment file, and from multiple files. ConfigMaps allow you to decouple configuration artifacts from image content to keep containerized applications portable.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
```

Step 2: Create a ConfigMap from Literal Values

Create a ConfigMap named example-configmap with literal values:

```
kubectl create configmap example-configmap --from-literal=key1=value1 --from-literal=key2=value2
```

Verify the ConfigMap:

```
kubectl get configmap example-configmap -o yaml
```

You should see the example-configmap with the keys and values specified.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create configmap example-configmap --from-literal=key1=value1 --from-literal=key2=value2
configmap/example-configmap created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get configmap example-configmap -o yaml
apiVersion: v1
data:
  key1: value1
  key2: value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T13:40:50Z"
  name: example-configmap
  namespace: default
  resourceVersion: "43650"
  uid: 694d632b-804d-4187-9f48-54427d341d47
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 3: Create a ConfigMap from a File

Create a file named **config.txt** with the following content:

```
key1=value1
key2=value2
```

Create a ConfigMap named file-configmap from the file:

```
kubectl create configmap file-configmap --from-file=config.txt
```

Verify the ConfigMap:

```
kubectl get configmap file-configmap -o yaml
```

You should see the file-configmap with the content of config.txt.

```
config.txt
key1=value1
key2=value2
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create configmap file-configmap --from-file=config.txt
configmap/file-configmap created
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get configmap file-configmap -o yaml
apiVersion: v1
data:
  config.txt: "key1=value1\r\nkey2=value2\r\n"
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T13:44:07Z"
  name: file-configmap
  namespace: default
  resourceVersion: "44008"
  uid: 27ca6c7a-450d-4f42-962e-d889f9dea974
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 4: Create a ConfigMap from an Environment File

Create an environment file named **env-config.env** with the following content:

```
ENV_VAR1=value1
ENV_VAR2=value2
```

Create a ConfigMap named env-configmap from the environment file:

```
kubectl create configmap env-configmap --from-env-file=env-config.env
```

Verify the ConfigMap:

```
kubectl get configmap env-configmap -o yaml
```

You should see the env-configmap with the environment variables specified in the file.

```
config.txt    env-config.env
env-config.env
ENV_VAR1=value1
ENV_VAR2=value2

# Add more environment variables as needed
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create configmap env-configmap --from-env-file=env-config.env
configmap/env-configmap created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get configmap env-configmap -o yaml
apiVersion: v1
data:
  ENV_VAR1: value1
  ENV_VAR2: value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T13:46:27Z"
  name: env-configmap
  namespace: default
  resourceVersion: "44265"
  uid: 0cd0b0cf-e8c1-41c8-aa80-1336cf7c3db9
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 5: Create a ConfigMap from Multiple Files

Create multiple files:

config1.txt with content:

```
key1=value1
```

config2.txt with content:

```
key2=value2
```

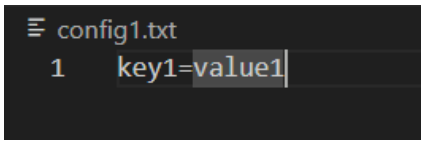
Create a ConfigMap named multi-file-configmap from multiple files:

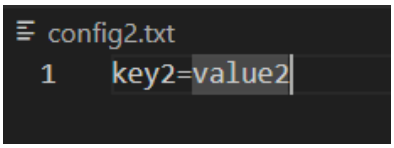
```
kubectl create configmap multi-file-configmap --from-file=config1.txt --from-file=config2.txt
```

Verify the ConfigMap:

```
kubectl get configmap multi-file-configmap -o yaml
```

You should see the multi-file-configmap with the content of both config1.txt and config2.txt.






```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create configmap multi-file-configmap --from-file=config1.txt --from-file=config2.txt
configmap/multi-file-configmap created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get configmap multi-file-configmap -o yaml
apiVersion: v1
data:
  config1.txt: key1=value1
  config2.txt: key2=value2
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T13:49:29Z"
  name: multi-file-configmap
  namespace: default
  resourceVersion: "44596"
  uid: e97ce9ca-8935-4049-98e3-176baf35b7b2
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 6: Use ConfigMaps in a Pod

Create a file named **pod-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: env-configmap
```

In this manifest:

- The Pod reads environment variables from the env-configmap.
- The Pod mounts the file-configmap at /etc/config.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-configmap.yaml
```

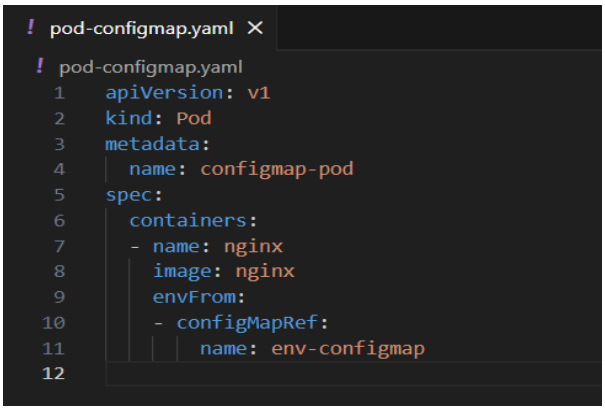
Check the status of the Pod:

```
kubectl get pods
```

View the Pod logs to see the environment variables and file content:

```
kubectl exec -it configmap-pod -- printenv
```

You should see the environment variables and the content of the file-configmap.



```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f pod-configmap.yaml
pod/configmap-pod created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME            READY   STATUS              RESTARTS   AGE
configmap-pod   0/1     ContainerCreating   0          8s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          18s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl exec -it configmap-pod -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
ENV_VAR2=value2
ENV_VAR1=value1
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
NGINX_VERSION=1.29.5
NJS_VERSION=0.9.5
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
HOME=/root
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 7: Clean Up

After completing the exercise, clean up the resources created:

kubectl delete pod configmap-pod

kubectl delete configmap example-configmap

kubectl delete configmap file-configmap

kubectl delete configmap env-configmap

kubectl delete configmap multi-file-configmap

rm config.txt env-config.env config1.txt config2.txt

For Poweshell :-

Remove-Item config.txt

Remove-Item env-config.env

Remove-Item config1.txt

Remove-Item config2.txt

Or

del config.txt

del env-config.env

del config1.txt

del config2.txt

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete pod configmap-pod
pod "configmap-pod" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete configmap example-configmap
configmap "example-configmap" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete configmap file-configmap
configmap "file-configmap" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete configmap env-configmap
configmap "env-configmap" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete configmap multi-file-configmap
configmap "multi-file-configmap" deleted from default namespace
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> wsl
root@Shivang:/mnt/c/Users/HP/Desktop/COAS_Lab/Lab_8# rm config.txt env-config.env config1.txt config2.txt
root@Shivang:/mnt/c/Users/HP/Desktop/COAS_Lab/Lab_8# exit
logout
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> dir

    Directory: C:\Users\HP\Desktop\COAS_Lab\Lab_8

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a---          25-02-2026    13:25          27664 Lab Exercise 15.docx
-a---          25-02-2026    13:25          26780 Lab Exercise 16.docx
-a---          25-02-2026    13:25          27666 Lab Exercise 17.docx
-a---          25-02-2026    13:25          28272 Lab Exercise 18.docx
-a---          28-02-2026    14:40         103792 Lab Exercise 19.docx
-a---          02-03-2026    19:22            186 pod-configmap.yaml

PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> ls

    Directory: C:\Users\HP\Desktop\COAS_Lab\Lab_8

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a---          25-02-2026    13:25          27664 Lab Exercise 15.docx
-a---          25-02-2026    13:25          26780 Lab Exercise 16.docx
-a---          25-02-2026    13:25          27666 Lab Exercise 17.docx
-a---          25-02-2026    13:25          28272 Lab Exercise 18.docx
-a---          28-02-2026    14:40         103792 Lab Exercise 19.docx
-a---          02-03-2026    19:22            186 pod-configmap.yaml

PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> Test-Path config.txt
False
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```